

Topic:

Find out the applications of all the inert gases.

Introduction:-

The inert gases also known as noble gases include helium He, neon Ne, argon Ar, krypton Kr, xenon Xe and radon Rn. These elements are called inert because they have a stable electron configuration, making them generally unreactive under normal conditions. Further by this project we will learn about its function or application.

◆ Here are some common uses of inert gases

☉ Lighting:-

Neon, argon and xenon gases are commonly used in various types of lighting applications. Neon gas produces the characteristic red orange glow in neon signs, while argon and xenon are used in fluorescent lamps, high intensity discharge HID lamps and plasma display panels.

☉ Welding:-

Argon and helium gases are often used as shielding gases in welding processes. They create an inert atmosphere around the weld, preventing the hot metal from reacting with atmospheric gases such as oxygen and nitrogen. This helps to produce high quality and clean welds.

☉ Filling balloons:-

Helium is widely used to fill balloons due to its low density and ability to make them float. It is commonly used in party balloons, scientific experiments and as a lifting gas for airships.

☉ Cryogenics:-

Helium is used extensively in cryogenic applications due to its extremely low boiling point. It is used to cool superconducting magnets in magnetic resonance imaging MRI machines, particle accelerators and other scientific equipment.

☉Calibration:-

Inert gases are used for calibrating gas detection instruments and sensors. Since these gases are chemically inert, they can be used to test the accuracy and sensitivity of gas detectors without interfering with the measurements.

☉Scuba diving mixtures:-

Helium is sometimes used in deep sea diving to prevent nitrogen narcosis, a condition that can occur at great depths due to increased pressure of nitrogen.

◆Here are some examples of uses of inert gases of each

☉Helium:-

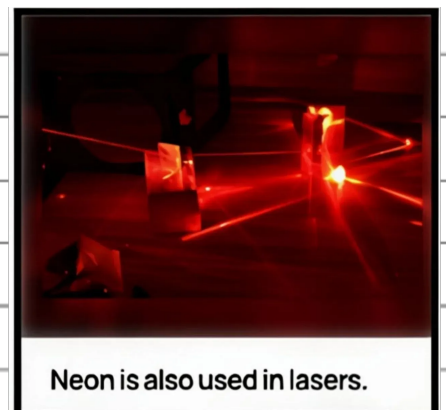


♣Used in balloons and blimps because it is lighter than air and does not burn

♣Used in gas chromatography as a carrier gas

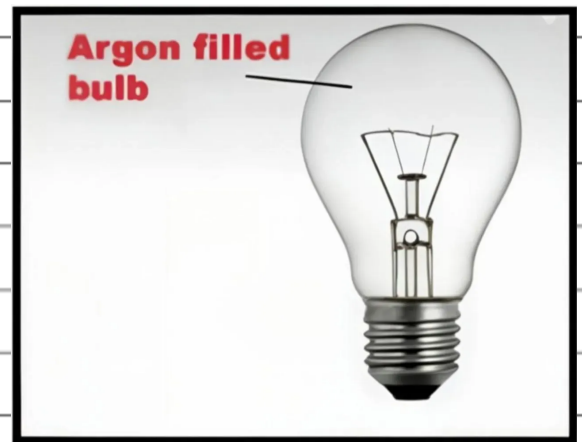
♣Used in cooling applications such as MRI machines and nuclear reactors

☉Neon:-



- ❖ Used in neon signs and lighting because of its bright orange red glow
- ❖ Used in gas lasers for scientific research

☉ Argon:-



- ❖ Used in welding to create an inert atmosphere around the weld and prevent oxidation
- ❖ Used in gas filled electric light bulbs to prevent the filament from oxidizing

☉ Radon:-



- ❖ Used in cancer radiation therapy to kill cancer cells
- ❖ Used in geological studies to help determine the age of rocks and minerals
- ❖ It is worth noting that while inert gases are generally unreactive, they can form compounds under certain conditions such as extreme heat or pressure
- ❖ These compounds are usually unstable and do not have many practical applications.

⊙Xenon :-



❖ Used in high-intensity lamps, such as those used in movie projectors and lighthouses.

❖ Used in anesthesia and surgery to produce a state of general anesthesia.

❖ Used in ion thrusters for spacecraft propulsion.

Advantages of inert gases:-

- Chemically stable and do not react easily under normal conditions
- Prevent oxidation and unwanted chemical reactions
- Non flammable and safe to use in industries
- Provide protective atmosphere in welding processes
- Widely used in lighting applications like neon signs and lamps
- Useful in medical and scientific equipment
- Help maintain clean and controlled environments

Disadvantages of inert gases:-

- ◆ Expensive to extract and store
- ◆ Require high pressure cylinders for storage
- ◆ Can cause suffocation in closed spaces by reducing oxygen level
- ◆ Some gases like radon are radioactive and harmful
- ◆ Need careful handling and safety precautions

Conclusion:-

Inert gases are essential in modern science and technology. Their non reactive nature makes them highly useful. They are widely used in lighting, welding, medicine and research . Advantages are more important than disadvantages. They play a key role in safe and efficient applications

References:-

- NCERT Science textbook
- School notes and classroom learning
- Educational chemistry websites.
- Encyclopedia Britannica.
- Online study resources..